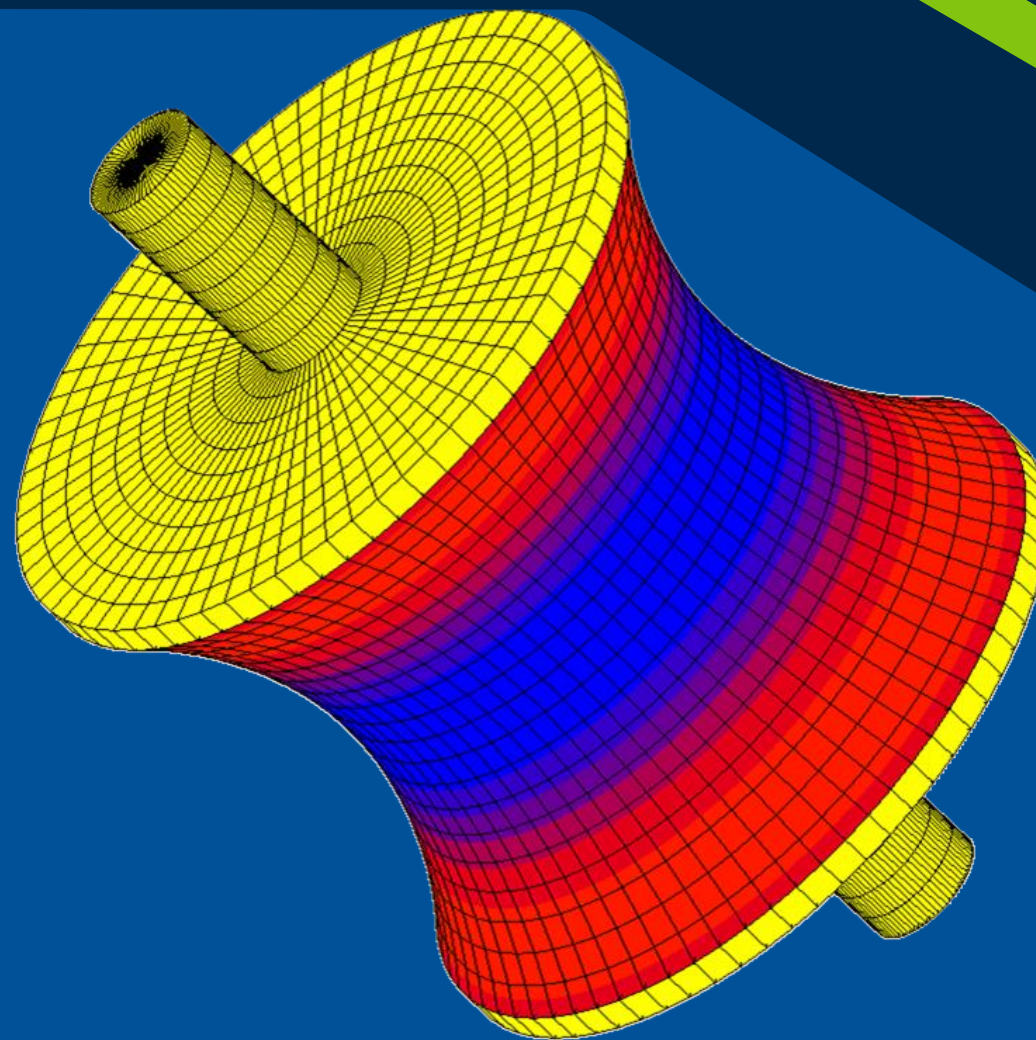




Marc

Release 2025.2:

What's new



Introduction to Marc

Marc | Advanced nonlinear simulation solution with 50+ years of development

- Marc is a structural simulation solution with strong multiphysics capabilities (thermal, thermomechanical, electromagnetics, piezoelectric, diffusion, etc.).
- The comprehensive nonlinear material model library makes Marc a market-leader in rubber and plastics modelling.
- Unique automatic remeshing capabilities make Marc a powerful nonlinear modelling solution.
- Marc excels at robust contact, failure, and damage analysis.
- Marc empowers advanced manufacturing process simulation.

The Value Proposition:
**A comprehensive, robust and
proven simulation solution for
complex engineering challenges
dealing with nonlinear
structural behaviours.**



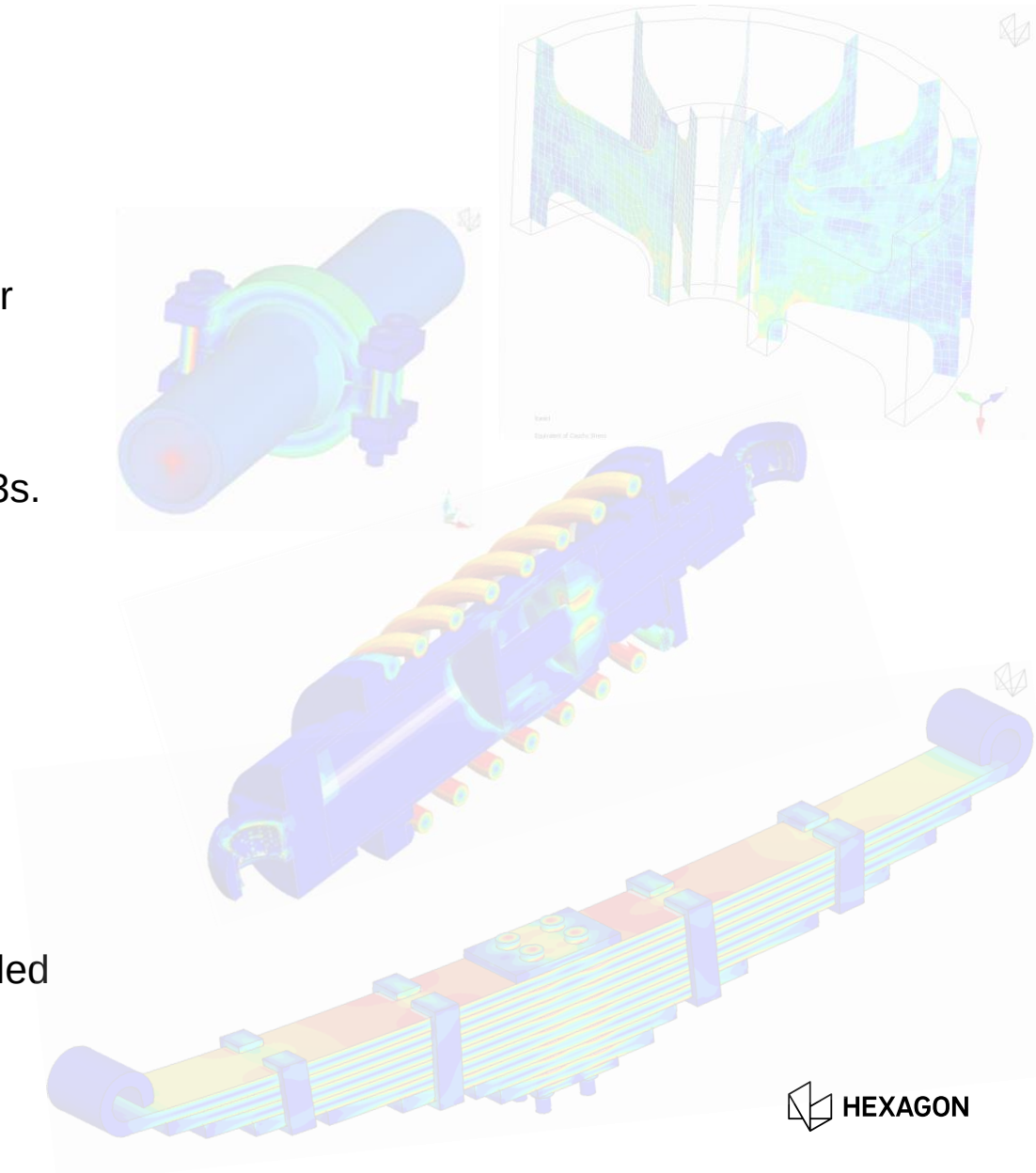
Release summary

Marc 2025.2

Release highlights

Major updates and enhancements in Marc 2025.2

- **Interpret results accurately:**
Mentat displays units consistently during postprocessing, whether you're viewing model plots, history plots, or path plots.
- **Take more control over RBEs:**
Easily show or hide, quickly identify, and select RBE2s and RBE3s. They are also visible during postprocessing.
- **Do more with the user material subroutine:**
The UMATRICAL subroutine now supports shells, beams, anisotropy, and tables. Reuse Abaqus UMAT subroutines with minimal or no changes to streamline workflows.
- **Take advantage of enhanced contact modelling:**
The hybrid node-to-segment contact method now enables user-defined separation behaviour, and glued contact with load-controlled rigid bodies results are more accurate.



New capabilities and enhancements in Marc 2025.2

User experience: Full units support for postprocessing

Description:

Marc 2025.2 introduces comprehensive unit awareness for postprocessing.

Marc writes the unit system of each new job to the results file. When performing analyses from a results file with units, Marc will display an error in the event of mismatched units to prevent inconsistencies. Mentat also automatically sets the model unit system to the unit system contained in the file. This system cannot be redefined or changed during post-processing.

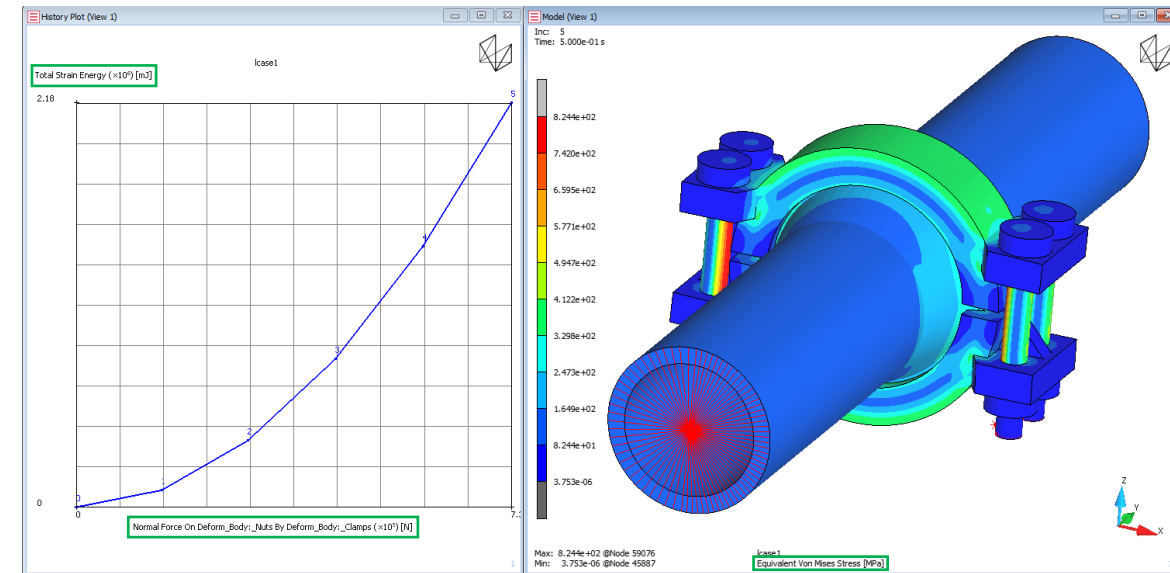
When opening older results files without units, users may define the model unit system. For model result plots (scalar, vector, tensor, beam diagram, curve distance, surface distance), Mentat displays applicable units in legends and headers. For path plot, history plot, and generalised XY-plots, Mentat displays units in both X and Y axes titles. Additional unit-related enhancements apply to tables, generalised XY-plots, the distance command, and several other commands.

Benefits:

- Results are clearer and easier to accurately interpret.
- Avoid confusion due to mismatched units

Who needs it:

- All users



History and model plots of a pipes connector model showing SI(mm) units

New capabilities and enhancements in Marc 2025.2

User experience: Enhancements with RBEs

Description:

Marc and Mentat now offer a significantly more intuitive and streamlined experience when working with RBE2 and RBE3.

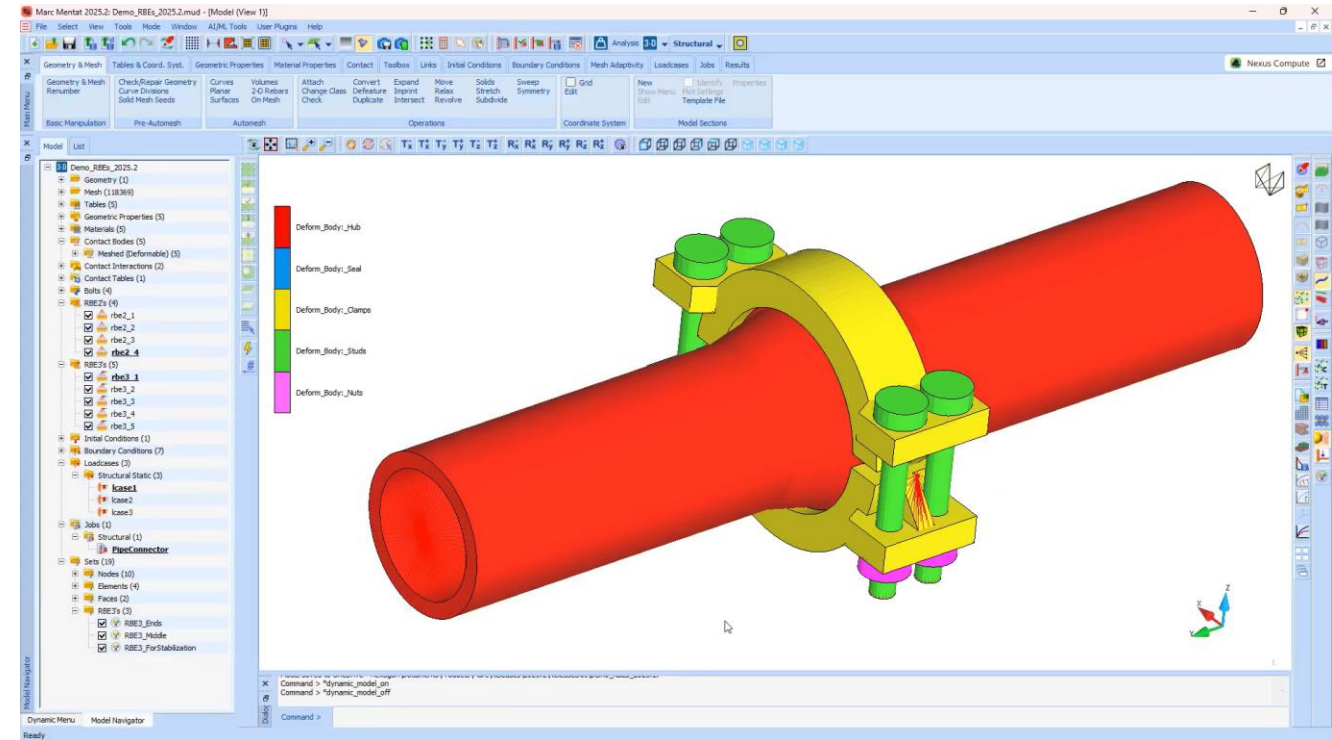
- Hide or show RBEs directly from the Mentat model navigator for better model visualisation.
- Select RBEs from the viewport using the selection control menu and manage visibility.
- Create RBE sets using the model navigator or the set control menu.
- Identify RBE sets through the model navigator's right-click menu or the Identification menu.
- Display RBEs during postprocessing. (Only supported for .t16 and .t19 result files, and for jobs that do not involve global remeshing or DDM.)

Benefits:

- Dealing with RBE2s and RBE3s in both pre- and post-processing is easier and more intuitive.
- Easily verify definitions during model creation, after job completion, or even while simulations are still running.

Who needs it:

- All users



RBE sets creation, visibility management, identification with a pipes' connector model

New capabilities and enhancements in Marc 2025.2

Capability enhancements: UMATERIAL subroutine

Description:

The user material framework introduced in Marc 2025.1 gives you more options and improves interoperability in version 2025.2. Now you can:

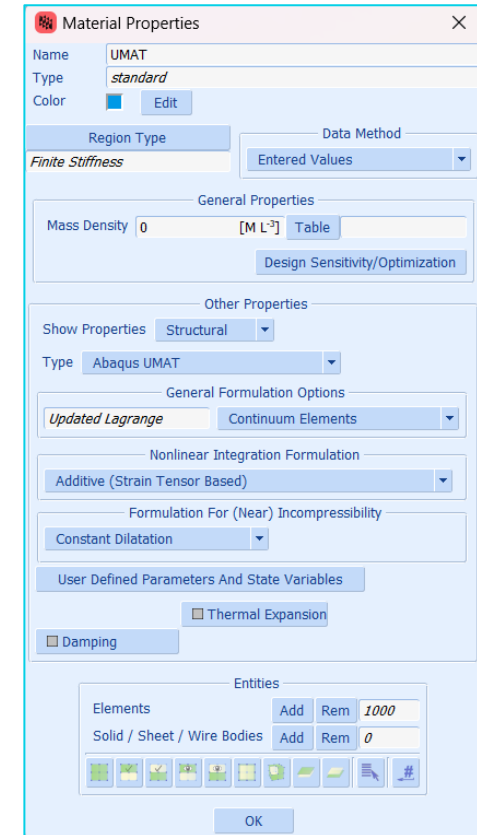
- Use tables for defining material properties variations with variables such as temperature or time.
- Use plane stress, beam, and truss formulations.
- Add material anisotropies and orientations.
- Reuse existing Abaqus UMAT subroutines for Marc analysis with the new external user material framework.

Benefits:

- Companies, research institutes, and universities that have already developed Abaqus UMAT subroutines for simulating advanced material behaviours can now use them with minimal to no modification in Marc analyses.
- Previous limitations of the UMATERIAL subroutine have been removed.

Who needs it:

- Users who need to define advanced user material models.



User defined material menu when the Abaqus UMAT type is selected

New capabilities and enhancements in Marc 2025.2

Capability enhancements: Contact modelling

Description:

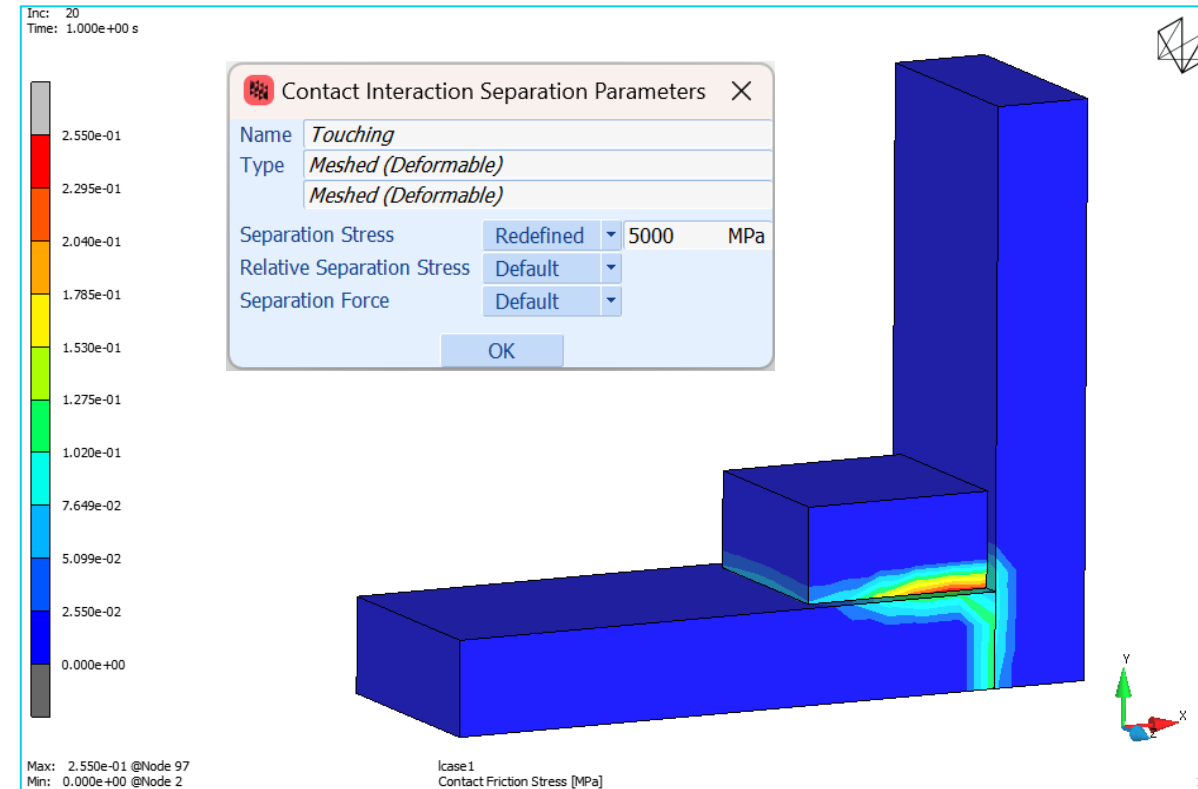
The hybrid node-to-segment contact method delivers greater robustness and versatility in contact simulations.

Benefits:

- Accurate handling of user-defined separation force or stress provides improved control over contact behaviour in specialised use cases.
- Glued contact with load-controlled rigid bodies enhances stability in complex models.
- Major improvements to stress-free initial contact and gap/overlap retention, particularly beneficial for analyses involving tetrahedral elements.

Who needs it:

- All users



Academic example where a user-defined separation stress is used with the hybrid node-to-segment contact method

New capabilities and enhancements in Marc 2025.2

Meshing capability enhancement: Imprint points on faces of solid or sheet bodies

Description:

The imprint option for adding a vertex to a solid body has been enhanced. Previously, points could only be imprinted on the edges of solids or sheet bodies. Now, you can also imprint points onto the faces of these bodies.

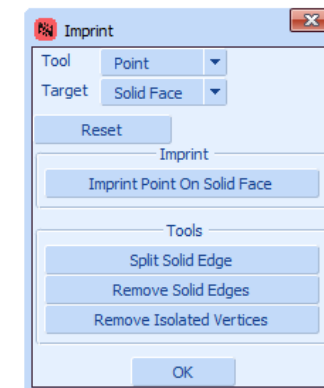
The point is projected onto the nearest location on the face, and a vertex is created at the projection point. When the body is meshed, a node is automatically created at the vertex. If a boundary condition is applied to the vertex, then it will be applied to that node for the simulations. The vertex can also have a mesh seed assigned, as shown in the figure on the right. Vertices imprinted on faces can be deleted if needed.

Benefits:

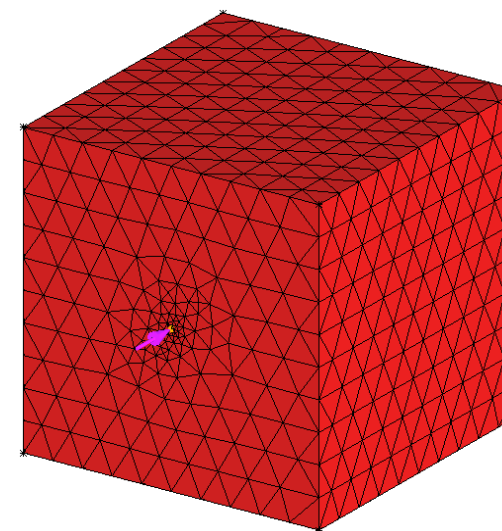
- Greater control and versatility for tuning solid meshes enables refined adjustments and precise node placement.

Who needs it:

- All users



Imprint menu



Point imprint option with mesh seed assignment used for a solid meshing



Previous releases

Marc 2024.1, Marc 2024.2 and Marc 2025.1

Highlights from Marc 2024.1 to 2025.1 releases

Hybrid contact formulation updates [2025.1]:

The double-sided contact algorithm has penalty-based enforcements for faster simulation run times and improved robustness and accuracy. This upgraded algorithm is now the default for new structural analyses.

New user material subroutine, UMATRICAL [2025.1]:

The UMATRICAL user material subroutine removes previous limitations and offers a more streamlined and flexible user experience.

Previous analysis state enhancements [2025.1]:

Use HDF5 files, map to larger models, and work with multiple result files.

New features and an improved user experience in Mentat [2025.1]:

Get deeper insight into material behaviour by viewing cutting planes in cylindrical coordinate systems, along with stress triaxiality output. The improved lighting option is enabled by default, the triad has been refreshed, and several other enhancements deliver a smoother and more intuitive user experience.

Fluid pressure penetration modelling [2024.2]:

The effect of the fluid seeping in under contact (e.g. a rubber seal) can now be modelled.

Hybrid contact enhancements [2024.1 and 2024.2]:

Contact detection improvements, reduction of iterations to enforce contact.

Units in Marc and Mentat in preprocessing [2024.1] :

The Mentat model now contains a complete unit system, fully defined by the user, to be used in pre-processing.

Temperature-dependent superelements [2024.1] :

Superelements can now have temperature dependencies for their structural behavior and can be exposed to 3rd party solution using a Functional Mock-Up Unit (FMU).

Hashiguchi advanced nonlinear friction model [2024.1] :

Gradual onset of nonlinear slip behavior and a smooth transition from static friction to a lower dynamic friction.

Parallel rheological framework data fitting [2024.1] :

The material support and data fitting for temperature dependent viscoelastic elastomer materials has been improved.



Additional resources

Marc 2025.2

Additional Resources



Support Email:
mscmarc.support@mscsoftware.com

Licensing Questions:
msc_lic.support@hexagon.com

Report a crash:
Use the Integrated Reporting Tool (IRT)

Click to visit
Nexus

Quick Access Links to Marc Resources

🕒 22 Feb, 2024 · Knowledge

Article Number

000023286

Description

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